Milestone 3 Report: Model Architecture, Baselines and Tuning

Project: Socratic AI Tutor for NCERT Classes 6–10
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Companion files: presentation (`M3_Presentation.pdf` / `.pptx`), work log (`Work_Contribution_Log.md`)
Models and dataset: https://huggingface.co/IITM-2026T3-DSAI-GROUP2

1. Summary

The tutor has three parts, and we chose and measured each one in M3:

Part	Job	Selected	Key result
Socratic response model	Write the next tutor turn as a guiding question, never the answer	Qwen 3.5-4B + QLoRA (r16, lr 2e-4)	Leak rate 0.01 → 0.00, reply length 56 → 20 words (tutors: 22), ROUGE-L 0.190 → 0.331 (+74%)
Knowledge tracing	Estimate per-skill mastery from the learner's answers	DKT (LSTM)	Test AUC 0.829 against 0.843 for the best model (AKT), with about a quarter of AKT's parameters and no need for question IDs
NCERT retrieval (RAG)	Ground every turn in the textbook	BM25 + BGE-small rank fusion over automatically built passages	Hit@5 80% on 15 held-out questions; 124 passages, 25 topics, 84 approved questions

A FastAPI demo app connects the three. The learner picks Class, Chapter and Topic, then chats with a tutor that cites NCERT passages, grades checkpoints and updates mastery after each answer. The fine-tuned adapters and the processed dialogue dataset are on the team's Hugging Face page.

2. Model architecture selection

2.1 Socratic response model

We fine-tuned four small open LLMs with QLoRA. Three of them share one protocol: the same data, the same 100 evaluation contexts and the same settings.

Candidate	Size	Architecture notes	LoRA targets	Licence
Qwen 3.5-4B (selected)	4B	Hybrid linear + full attention; thinking mode off	q, k, v, o, gate, up, down	Apache 2.0
Gemma 4 E4B	~4B effective	Multimodal base; vision and audio towers excluded from LoRA	language-model q… down	Gemma terms
Phi-4-mini-instruct	3.8B	Phi-3 family, fused qkv and gate_up projections	qkv, o, gate_up, down	MIT
Qwen3-4B-Instruct-2507	4B	Early feasibility trial on a different split; loss only, not ranked	q…down (α 32)	Apache 2.0

2.2 Knowledge tracing

Seven models from three families were trained on ASSISTments 2009 with one shared training engine: DKT with LSTM, GRU and plain RNN cells [8]; SAKT with and without positional embeddings [9]; and AKT with and without Rasch question difficulty [10]. DKT keeps a single recurrent knowledge state. SAKT attends over past answers. AKT adds a learned decay on old practice and, in the Rasch version, a difficulty term for each individual question.

2.3 Retrieval

We compared three retrievers on a frozen set of textbook questions: BM25 keyword search, BGE-small-en-v1.5 dense embeddings (384-d) [12], and reciprocal rank fusion of the two. The pipeline stores passages in PostgreSQL + pgvector; the demo app loads them from a frozen file.

3. Architecture justification

3.1 Why a fine-tuned small LLM

Untrained models already know the maths; what they lack is tutoring behaviour. They solve and explain at 2 to 2.5 times a real tutor's length, and Phi-4-mini states the answer in 6% of replies (12% on MathDial). Fine-tuning on real tutoring dialogues targets that behaviour directly.

The literature points the same way. In MathDial, a fine-tuned Flan-T5 (780M) told students the answer in under 1% of turns (Telling@5), against 16% for prompted ChatGPT [3]. SocraticLM, a fine-tuned 6B model, beat GPT-4 on ROUGE-L (47.5 vs 32.4) on SocraTeach [4].

A 4B model in 4-bit also trains on one free 16 GB Kaggle T4 and runs on infrastructure the team hosts, so student conversations do not go to a paid third-party API.

Qwen 3.5-4B was chosen over the alternatives on the tuned results. It has the best overall ROUGE-L (0.331 vs 0.319 for Gemma and 0.294 for Phi), the best MathDial ROUGE-L (0.244 vs 0.228 and 0.186), zero leaks, and a MathDial question rate (0.42) closest to real teachers (0.44). It is Apache 2.0, and its adapter is already published and served to the app. Gemma is close enough to call a tie on 100 contexts and one seed. Phi trains fastest (49 min) but falls behind on MathDial.

3.2 Why DKT for the app

AKT with Rasch difficulty has the best test AUC (0.843), 0.014 ahead of DKT (GRU) and 0.015 ahead of DKT (LSTM). That lead comes from per-question difficulty, which needs ASSISTments question IDs. NCERT questions have no such IDs, and without the Rasch term AKT scores 0.826, below both DKT variants. DKT (LSTM) has 131K parameters against AKT's 491K and updates in one step per answer. With a single seed, the GRU and LSTM results (0.8291 and 0.8286) are a tie.

3.3 Design decisions

- Loss is computed on tutor turns only. Turns flagged as leaking or "telling" stay in the context but are never targets.
- The correct answer is in the system prompt so the model can check the learner's work, and we measure the leak rate on every run.
- The end-of-turn token is trained so the model stops after one turn.
- QLoRA uses a 4-bit NF4 base and fp16 compute, with 10 to 35M trainable parameters [1, 2].
- Knowledge-tracing data is split by student, so test students are never seen in training.

3.4 Trade-offs

Choice	Gain	Cost
4B model, not 7–8B	Trains and serves on a free T4	Probably a lower ceiling on reasoning

Choice	Gain	Cost
768-token cap	Fits T4 memory	About 22% of dialogues are cut back to the last tutor turn that fits
fp16 (T4 has no bf16)	Runs on free hardware	Needs a finite-loss check before training
ROUGE/BLEU	Cheap and reproducible	They reward wording, not teaching, so we report leak rate, question rate and length beside them
DKT over AKT	Works with skill IDs alone; 4× fewer parameters	Gives up 0.015 AUC on ASSISTments
Learner picks the topic	No grounding in the wrong topic	One extra click; automatic routing failed its test (Section 6.3)

4. Baseline performance and candidate dataset

4.1 Candidate dataset

The dialogues come from SocraTeach (LLM-generated Socratic dialogues) and MathDial (real teachers correcting student errors). Full detail is in `notes/hariom kumar/milestone-3/socratic_data_prep/REPORT.md`, and the processed data is on Hugging Face.

Split	SocraTeach	MathDial	Use
Full train	22,574	1,302	Source pool
Full validation	2,682	311	Source pool
`candidate_train`	1,000	1,000	Grid search (the same 400-dialogue subset for every model) and final training
`candidate_val`	100	100	100 evaluation contexts (50 per source) and validation loss
MathDial test	0	599	Untouched, kept for the final evaluation in M4

The split is by problem, using a `sha1` of the normalised question. Every train dialogue that shares a problem with the test set is removed, and the notebook asserts that no problem appears in two splits. Within each source, the subset is stratified by word-count quartile with seed 42.

A two-sample KS test compares the subset with the full train split:

Source	Feature	Full train	Subset	KS	p
SocraTeach	Words per dialogue	234.9	234.3	0.020	0.81
SocraTeach	Messages per dialogue	9.71	9.69	0.013	0.99
MathDial	Words per dialogue	314.2	312.9	0.010	1.00
MathDial	Messages per dialogue	12.04	12.03	0.013	1.00
MathDial	Leak share of tutor turns	0.017	0.017	0.004	1.00
MathDial	“Telling” share of tutor turns	0.154	0.153	0.006	1.00

MathDial is 5.5% of full train but 50% of the subset. We did this on purpose: MathDial is the only real-teacher data and makes up the whole test set, and at its natural share the subset would hold about 110 MathDial dialogues. Results are therefore reported per source.

For knowledge tracing we use the ASSISTments 2009 skill-builder data: 123 skills and 15,922 questions (`skill_map.csv`, `question_map.csv`). Rows without a skill and scaffolding sub-questions are dropped, students are split 80/10/10, and the test split gives 27,329 predictions on students the models never saw.

4.2 Metrics

Metric	Definition	Target
Leak rate	Reply states the final numeric answer before the student has	Lower; real tutors 0.01
Question rate	Reply contains a question	Near real tutors (0.62 overall)
Words	Reply length	Near real tutors (22.2)
ROUGE-L [6]	Longest-common-subsequence overlap with the real tutor's next turn	Higher
BLEU [7]	sacreBLEU ÷ 100 against the real next turn	Higher
Validation loss	Cross-entropy on tutor turns	Compared within a model only, since tokenizers differ
KT: AUC, accuracy, F1	Correctness of the next answer	Higher; chance AUC is 0.5
RAG: Hit@1, Hit@5, MRR	Rank of the gold passage among results	Higher

4.3 Socratic baselines

Pretrained weights in 4-bit, greedy decoding, at most 96 new tokens, on the same 100 contexts:

Baseline	Leak	Question	Words	ROUGE-L	BLEU	MathDial ROUGE-L
Real tutors	0.01	0.62	22.2	n/a	n/a	n/a
Gemma 4 E4B	0.00	0.87	48.0	0.189	0.040	0.111
Qwen 3.5-4B	0.01	0.52	56.4	0.190	0.040	0.124
Phi-4-mini	0.06	0.60	39.6	0.226	0.071	0.137

The knowledge-tracing baseline always predicts the majority answer ("correct"): accuracy 0.670, AUC 0.500.

4.4 Published benchmarks

These use different splits and metrics, so they put our numbers in context rather than rank them.

Source	Setting	Result
MathDial [3]	Flan-T5 780M fine-tuned, MathDial test	sBLEU 9.7; Telling@5 <1% vs ChatGPT 16%
SocraticLM [4]	ChatGLM3-6B on SocraTeach vs GPT-4	ROUGE-L 47.5 vs 32.4; BLEU-4 48.6 vs 36.2
DKT [8] and re-check [11]	ASSISTments 2009	AUC about 0.86 as first reported; about 0.75 after duplicate rows were removed
Ours, tuned Qwen	50 SocraTeach / 50 MathDial contexts	ROUGE-L 41.9 / 24.4
Ours, KT	Student split, one row per (order, skill)	AUC 0.826 to 0.843 for DKT and AKT

Our tuned SocraTeach ROUGE-L sits between GPT-4 and SocraticLM's fine-tuned 6B model. Our KT preprocessing keeps a problem tagged with several skills as several rows with the same answer, the effect [11] identified, so our

AUCs are likely optimistic in absolute terms. The comparison between our seven models is still fair because they all see the same data.

5. Hyperparameter tuning

5.1 Socratic model

We ran a grid search of 3 settings for each of 3 models. Each setting trains for 1 epoch on the same 400 candidate dialogues (about 50 optimizer steps) and is scored on the same 100 contexts. We fixed the selection rule before training: pick the lowest validation loss among settings whose leak rate is at most 0.02 above real tutors.

Hyperparameter	Tried	Rationale
LoRA rank r (α = r)	8, 16	Tests whether more adapter capacity helps; α = r keeps the scale at 1
Learning rate	2e-4, 5e-5	2e-4 is the QLoRA default [2]; 5e-5 tests a gentler update
Fixed	n/a	dropout 0.05, cosine schedule with 5% warm-up, paged AdamW 8-bit, effective batch 8, max 768 tokens, seed 42

Grid results (ROUGE-L / validation loss / leak):

Model	r8, 2e-4	r16, 2e-4 (selected)	r8, 5e-5
Gemma 4 E4B	0.250 / 1.260 / 0.02	0.267 / 1.233 / 0.00	0.216 / 1.406 / 0.03
Qwen 3.5-4B	0.302 / 1.169 / 0.00	0.308 / 1.154 / 0.00	0.259 / 1.255 / 0.01
Phi-4-mini	0.265 / 1.273 / 0.01	0.266 / 1.256 / 0.01	0.220 / 1.403 / 0.00

The low learning rate under-trains every model: it gives the highest loss and lowest ROUGE-L each time. At 2e-4, rank 16 beats rank 8 for all three at the same 10 to 19 minute cost. We then trained r16 / 2e-4 on the full 2,000-dialogue candidate set.

5.2 Tuned vs baseline

Final model	Leak	Question (0.62)	Words (22.2)	ROUGE-L	BLEU	Val loss	Train time
Gemma 4 E4B	0.00	0.69	20.5	0.319	0.175	2.76 → 1.12	62 min
Qwen 3.5-4B	0.00	0.59	20.3	0.331	0.176	1.82 → 1.07	80 min
Phi-4-mini	0.00	0.65	21.8	0.294	0.174	1.85 → 1.17	49 min

Per source (50 each)	SocraTeach ROUGE-L	MathDial ROUGE-L	SocraTeach Q (0.80)	MathDial Q (0.44)
Gemma 4 E4B	0.409	0.228	0.80	0.58
Qwen 3.5-4B	0.419	0.244	0.76	0.42
Phi-4-mini	0.402	0.186	0.80	0.50

All three now write replies of human length with zero leaks. ROUGE-L rose 69% for Gemma, 74% for Qwen and 30% for Phi. The comparison plots are in `notes/hariom kumar/milestone-3/model_finetune/*_finetune_output/` and on slides 7 and 8.

5.3 Knowledge tracing

KT used no per-model search. Every model gets the same defaults so the ranking compares architectures: Adam at lr 1e-3, batch 64, gradient-norm clip 5, 200-step windows, seed 42, and up to 60 epochs with early stopping after 8 epochs without a gain in validation AUC. The epoch with the best validation AUC is restored, and the test set is scored once.

Model	Params	Best epoch	Val AUC	Test AUC	Accuracy	F1
DKT (LSTM), selected	131,003	11	0.826	0.829	0.780	0.846
DKT (GRU)	106,171	14	0.830	0.829	0.778	0.845
DKT (RNN)	56,507	20	0.825	0.826	0.777	0.844
SAKT	49,025	30	0.767	0.765	0.744	0.827
SAKT + positions	61,825	27	0.801	0.805	0.767	0.842
AKT (no Rasch)	428,057	25	0.828	0.826	0.776	0.845
AKT (Rasch)	491,468	16	0.836	0.843	0.787	0.848
Majority class	0	n/a	n/a	0.500	0.670	n/a

All three DKT cells land within 0.003 AUC of each other. Positional embeddings add 0.040 AUC to SAKT. The Rasch difficulty term adds 0.017 AUC to AKT, the largest single gain in the table. Full results are in `KT_model/results_summary.csv`, and the AKT test predictions are in `akt_test_predictions.csv`.

5.4 Retrieval

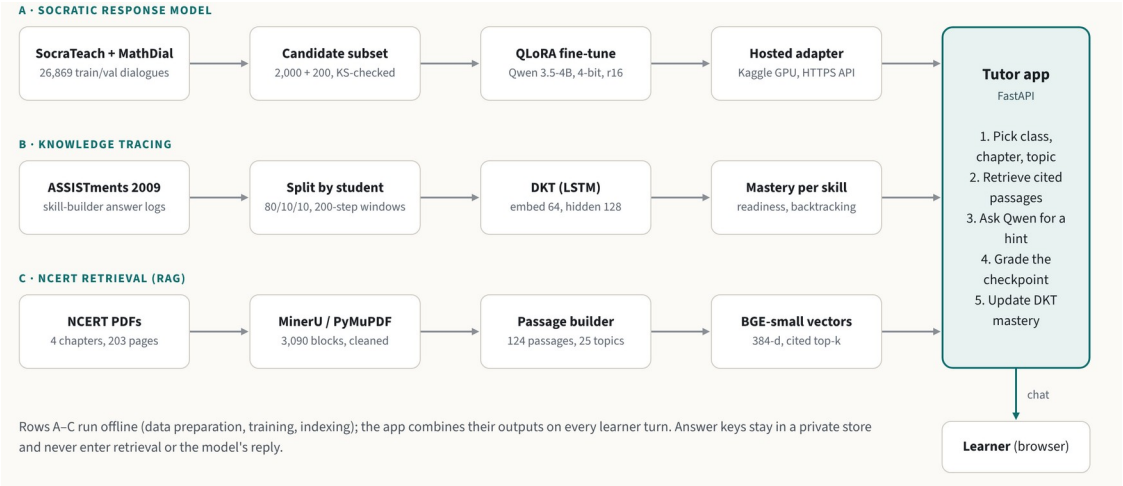
Retrieval has no trained weights. The tuned values are the passage size (300-token target, 400 maximum, 25 minimum) and the routing cut-offs (similarity ≥ 0.90, margin ≥ 0.10). We set them on 24 calibration questions and froze them before the held-out test.

Retriever (15 held-out)	Hit@1	Hit@5	MRR
BM25	53.3%	73.3%	0.597
BGE-small	46.7%	73.3%	0.572
Rank fusion (used)	46.7%	80.0%	0.591

The labels cover only the six original pilot concepts, so some misses land on new topics that are actually correct. We did not change the labels after seeing the results. For misconception retrieval on 300 Eedi groups, Recall@5 rose from 16% (BGE) to 26% (contrast query + BM25). That is too weak to label learners, so the app shows these suggestions as unconfirmed.

6. End-to-end setup

6.1 Architecture diagram



End-to-end architecture. Rows A to C run offline (data preparation, training, indexing); the tutor app combines their outputs on every learner turn.

6.2 Data flow, training, validation, evaluation and outputs

Stage	Socratic model	Knowledge tracing	NCERT retrieval
Input data	SocraTeach + MathDial → chat format, loss on tutor turns, 768 tokens	ASSISTments 2009 → per-student sequences, 200-step windows	9 pinned NCERT PDFs → passages, questions, private keys
Training	QLoRA on Kaggle T4; grid on 400 dialogues → final r16 on 2,000	7 architectures, shared defaults, up to 60 epochs with early stopping	None: BGE-small frozen; BM25 index; calibrated cut-offs
Validation	Val loss each run; 100 fixed contexts, 50 per source	Val AUC each epoch; best epoch restored	24 calibration questions; release gates and fingerprints
Evaluation	Leak, question rate, words, ROUGE-L, BLEU vs baseline	Test AUC, accuracy, F1, scored once	Hit@1, Hit@5, MRR on 15 held-out; Eedi Recall@5
Expected output	One short Socratic turn (about 20 words) that does not state the answer	Per-skill probability of a correct next answer	Cited passages ("book · PDF page") and a gradable checkpoint

6.3 NCERT pipeline

1. Parse: MinerU handles maths and answer pages (74 of 75 maths pages contain formulas); PyMuPDF handles plain science pages.
2. Clean: drop page numbers, running heads, question boxes, teacher notes, figures, tables, unclear ₹ amounts and broken markup.
3. Passages: an automatic builder merges blocks by numbered section heading (300-token target, 400 maximum). Section headings become topics, so the same code runs for Classes 6–10 without manual chunking.
4. Questions: chapter questions are matched to answer-page keys by printed page. The 16 questions that depend on a figure are skipped until the app can show figures, and unmatched keys are reported instead of being turned into questions.
5. Index and release: BGE-small vectors in pgvector plus BM25, frozen into a fingerprinted release (`ncert-pilot-a2ac2147befaf6e1`). Answer keys live in a private schema and never enter a prompt.

At the calibrated cut-offs, automatic topic routing accepted 0 of 15 answerable questions, so it is disabled. The learner picks the topic and retrieval searches only inside it.

6.4 Demo app status

The FastAPI app keeps learner history in SQLite and offers Class → Chapter → Topic navigation and three hint levels. It grades checkpoints against private keys, tracks readiness with DKT and backtracks to prerequisites, screens replies for leaks and truncation, and keeps progress when a model call fails. Unit tests and `check_offline.py` run on the real release with zero model calls; 52 tests pass across the pipeline and app. The Qwen adapter is hosted behind an HTTPS tunnel from a Kaggle or Colab notebook (`notes/hariom kumar/milestone-3/model_hosting/README.md`).

Two things are not yet tested. The live Qwen step with retrieved passages (`check_live.py`) is scheduled for M4. DKT is trained on ASSISTments skills, so only NCERT topics that map to one of them get a model prediction; the others use practice estimates.

7. Limitations and next steps

Limitation	Plan for M4
100 contexts per model and one seed; Qwen vs Gemma is a near tie	Score all models once on the 599-dialogue MathDial test split, with Telling@k as in [3]
ROUGE/BLEU reward wording, not teaching	Add a rubric rating from a human sample and an LLM judge
The tutoring data is English maths word problems	Evaluate on NCERT science topics; consider NCERT-grounded dialogues
KT is one seed, untuned, with multi-skill rows kept and no BKT baseline	Run 3 to 5 seeds, add BKT, retrain without duplicate rows, fit or calibrate on in-app answers
15 held-out retrieval questions, labels for 6 of 25 topics	Extend the builder to all Classes 6–10 books, label a fresh test against every topic, re-test routing
The leak guard is lexical	Run `check_live.py` end to end and review a sample of live replies

8. Contributions

Member	M3 work
Sandeep Nalla	Zero-shot baseline evaluation script (ROUGE-L, question rate, reply length, per source); Phi-4-mini-instruct QLoRA grid and final run; Phi adapter on Hugging Face
Hariom Kumar	Team coordination; Socratic data preparation and candidate set; Gemma 4 E4B and Qwen 3.5-4B fine-tuning and the four-model comparison; NCERT pilot; demo app; Qwen hosting; dataset and model uploads to Hugging Face; this report and the presentation
Hariom Pandey	Knowledge-tracing comparison of DKT, SAKT and AKT (with Mrityunjay Chakraborty); Qwen3-4B-Instruct fine-tuning trial (`claudeaisocratic.ipynb`)
Mrityunjay Chakraborty	Knowledge-tracing comparison of DKT, SAKT and AKT (with Hariom Pandey)

9. Evidence

Area	Location
Data preparation	`notes/hariom kumar/milestone-3/socratic_data_prep/REPORT.md`
Socratic fine-tuning (4 models)	`notes/hariom kumar/milestone-3/model_finetune/report.md`, `README.md`, notebooks and `*_output/results.json`
Models and dataset	https://huggingface.co/IITM-2026T3-DSAI-GROUP2
Model hosting	`notes/hariom kumar/milestone-3/model_hosting/README.md`

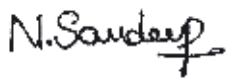
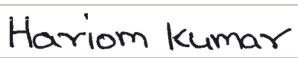
Area	Location
Knowledge tracing	`docs/milestone-3/KT_model/`: `KT_Models_Comparison_AKT_firstnb_processing.ipynb`, `results_summary.csv`, `skill_map.csv`, `question_map.csv`, `akt_test_predictions.csv`
NCERT pipeline	`Ncert work/PILOT_REPORT.md`, `Ncert work/README.md`
Demo app	`mock-app/README.md`, `mock-app/VERIFICATION.md`
Presentation	Design canvas exported as `M3_Presentation.pdf` (13 slides)

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Sign-off

By signing, each member confirms they have reviewed this report, the presentation and the work log.

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